

Improving the Accuracy of SSPA Device Behavioral Modeling

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Abstract — Several well-known simple behavioral models for solid state power amplifier (SSPA) devices are reviewed and compared. This paper proposes an improvement to White et al's model and discusses their use of the Rapp model for AM/AM and AM/PM device modeling. Furthermore an improvement to Honkanen & Haggman's phase shift addition to Rapp's model is given thus allowing phase shifts greater than zero degrees to be considered. Finally a review of several models' performances beyond available data and into the saturation region is presented.

These simple model improvements and additions, combined with the extension of the modeling into the saturation region, thus allow for faster and more accurate system level analysis, enabling better crucial output device selections to be made.

Index Terms — Behavioral models, SSPAs, White Model, Rapp Model.

I. INTRODUCTION

In order to quickly compare the suitability of different RF power amplifier (RFPA) devices, a system level analysis needs to be performed. Typically RFPA devices are the primary source of nonlinearities in multicarrier systems, so selecting the right device is crucial. To quickly compare RFPA devices within a system, accurate and simple behavioral models are needed.

Obtaining a RFPA device behavioral model requires either custom testing, which can be very time consuming, or manufacturer's datasheets can be used. Datasheets however, may not always provide information or data for applications where an RFPA device will be used, for example when an RFPA device is used deep into saturation, associated with modern digital modulation schemes, the peak to average power ratios (PARPs) can range from 3 to 10 dB. Therefore a model that accurately predicts the performance of an RFPA device, over its' full range of operating conditions, using parameters extracted from manufacturer's limited data, is highly desirable.

This paper reviews several behavioral models commonly adopted in the literature, Section II. In addition, in Section III, it compares their applicability as system behavioral models when parameters for these models are extracted from manufacturer's limited data. Device modeling in the saturation region is presented in Section IV, followed by an amplitude/amplitude

(AM/AM) and amplitude/phase modulation (AM/PM) modeling discussion in Section V. These are then followed by a conclusion in Section VI and an acknowledgment.

II. REVIEW OF EXISTING BEHAVIORAL MODELS

Saleh [1] presented a Traveling Wave Tube Amplifier (TWTA) model that does not model SSPAs well. Ghorbani & Sheikhan [2] proposed an SSPA model with both the Saleh and Ghorbani & Sheikhan models including AM/AM and AM/PM terms. Rapp [3] proposed another SSPA model providing a better saturation performance but only for AM/AM, as Rapp suggested that AM/PM conversion is small enough to be neglected. Honkanen & Haggman [4] provided an AM/PM expression for the Rapp model. White et al [5] indicated an improved accuracy model compared to [1], [2], and [3]. An improved accuracy model for a laterally diffused metal oxide semiconductor (LDMOS) based SSPA was proposed by O'Dorma et al [6]. A summary of these models is given in Table 1.

Table 1
Simple behavioral models for RF power amplifiers.

Model	AM/AM	AM/PM
Saleh [1]	$A_S(r) = \frac{\alpha_a r}{1 + \beta_a r^2}$	$\phi_s(r) = \frac{\alpha_\phi r^2}{1 + \beta_\phi r^2}$
Ghorbani & Sheikhan [2]	$A_G(r) = \frac{x_1 r^{x_2}}{1 + x_3 r^{x_2}} + x_4 r$	$\phi_G(r) = \frac{y_1 r^{y_2}}{1 + y_3 r^{y_2}} + y_4 r$
Rapp [3] (with Honkanen & Haggman [4] Phase)	$A_R(r) = \frac{A \times r}{(1 + (A \times r / \alpha)^{2\beta})^{1/2\beta}}$	$\phi_{RH}(r) = \frac{b}{1 + e^{-c(r-a)}} \cdot \frac{1}{p_1 r + p_2}$
White et al [5]	$A_W(r) = a(1 - e^{-br}) + cre^{-dr^2}$	$\phi_W(r) = f(1 - e^{-g(r-h)}), r \geq h$ $\phi_W(r) = 0, r < h$
O'Dorma et al [6]	$A_{MS}(r) = \frac{\alpha_a r^\eta}{(1 + \beta_a r^\gamma)^\nu} - \varepsilon$	$\phi_{MS}(r) = \frac{\alpha_\phi r^\eta}{(1 + \beta_\phi r^\gamma)^\nu} - \varepsilon$

III. EXISTING MODEL IMPROVEMENTS

White et al's [5] original AM/AM equation is presented in Table 1, with $A_W(r)$ being the output amplitude and r the normalized input amplitude. Coefficients a and b determine the saturation level and small signal gain respectively while parameters c & d adjust the shape in the SSPA operating region.

Attempts at reproducing White et al's [5] results, using this equation give the results shown in Fig. 1a. These results do not match the published results of White et al's Fig. 1b [5]. It has been found that to reproduce the correct results, as proposed by White et al's Fig. 1b [5] then the AM/AM model, $A_{WC}(r)$ should be written as;

$$A_{WC}(r) = a[(1 - e^{-br}) + cre^{-dr^2}] \quad (1)$$

Using the corrected equation (1), the error curves, as shown in Fig. 1b, now match White's original Fig. 1b curve.

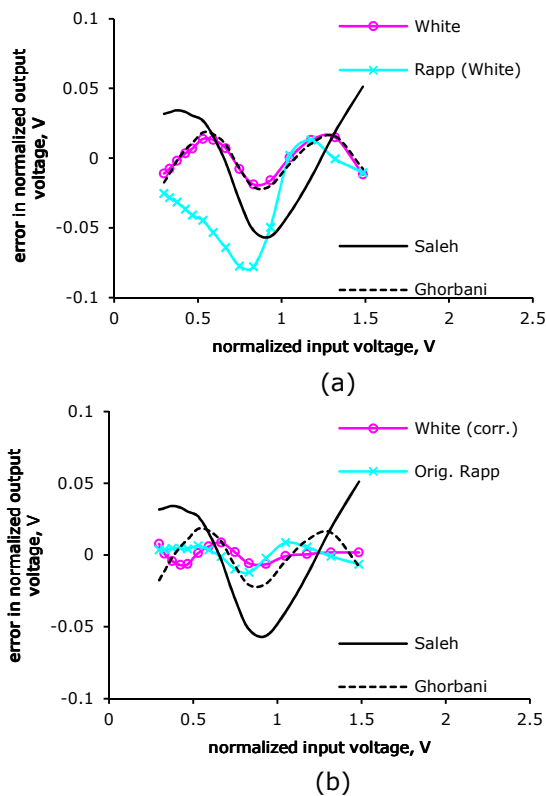


Fig. 1 Curve fit of models showing the AM/AM normalized output voltage errors for the measured data dynamic range only. (a) White et al's Fig 1b using the equation as given in that paper. (b) Corrected White et al equation (1) & original Rapp equation, Table 1.

Also within White et al's paper, Rapp's equation is given as;

$$A_{RW}(r) = r/(1 + (|r|/\alpha)^{2\beta})^{1/2\beta} \quad (2)$$

However if Rapp's original equation, as per Table 1, is used, including the small signal gain parameter A , then the error results are as shown in Fig. 1b and the AM/AM curve fit is shown in Fig. 3, indicating that Rapp's equation now has similar error performance compared to White et al's equation. An AM/AM normalized mean squared error (NMSE) figure-of-merit (FOM) comparison, as defined in [7], is given in Table 2 indicating that the original Rapp model compares well with White et al's corrected model.

Table 2
NMSE FOM results for AM/AM Modelling.

Model	AM/AM NMSE (dB)
Saleh [1]	-27.13
Ghorbani & Sheikhan [2]	-35.44
Rapp [3]	-42.33
Rapp from White et al [5]	-24.97
White et al [5]	-36.95
White et al, corrected (this paper)	-44.00
Modified Saleh, O'Dorma et al [6]	-48.54

IV. MODELING INTO THE SATURATION REGION

In order to compare the original Rapp, White et al (corrected) and O'Dorma et al models, beyond the available data, a method is used to extend the data into the saturation region. This method is to observe that the maximum end point output power (P_{out}), of the power added efficiency (PAE) versus P_{out} curve(s), typically provided for RFPA devices, is the final saturated output power, Fig. 2.

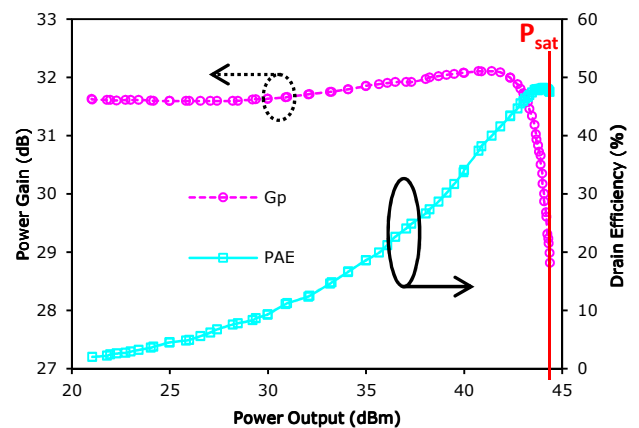


Fig. 2 Power gain & Drain efficiency for the BLM7G1822S [8] providing the saturated output power.

The existing data can then be extended into the saturation region.

Extending the modeling beyond the measured data and into the saturation region, the AM/AM curves of Figs. 3 &

4 show how the models perform beyond the available data. It can be seen that the original White et al equation 'rolls over' similar to the Ghorbani & Sheikhan model [2]. If the corrected White et al equation (1) is used, then it performs as would be expected for an LDMOS based SSPA.

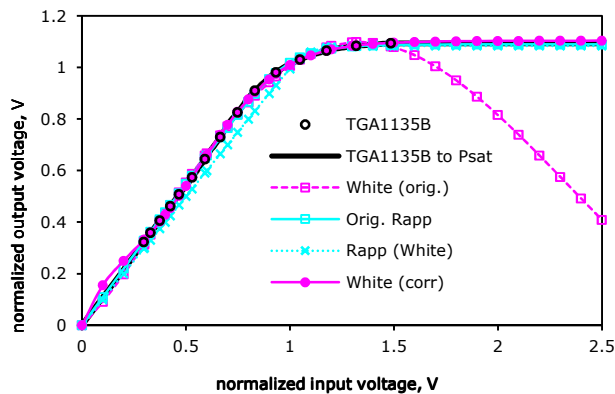


Fig. 3 Measured AM/AM data and curve fit of normalized output voltage versus normalized input voltage into the saturation region for White et al's original and corrected equation comparisons. Also shown is Rapp's original equation and Rapp's equation (2) as per White et al.

Results indicate that the Rapp model performs best into the saturation region when only available data is used to curve fit the models, even compared to the O'Dorma et al model [6] as shown in Fig. 4.

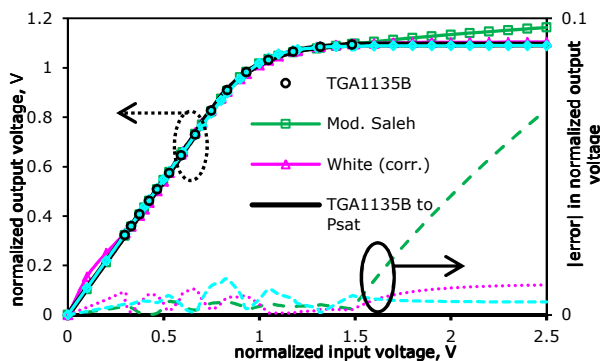


Fig. 4 Rapp, White et al (corrected) & O'Dorma et al model comparison of AM/AM curve fit showing normalized output vs input voltage and absolute normalized output voltage errors for existing data and data extended into the saturation region.

When the extended data is used to curve fit the model then both the Rapp & O'Dorma et al models are comparable to the corrected White et al model, as indicated in Fig. 5.

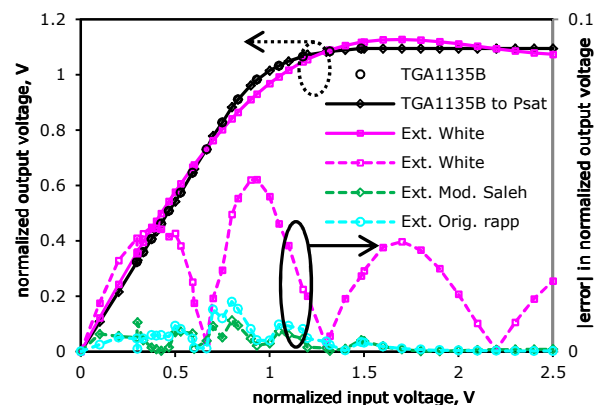


Fig. 5 Corrected White et al AM/AM model Curve fit of extended data into the P_{sat} region showing that it performs worse than the original Rapp or O'Dorma et al models.

V. AM/PM

The Honkanen & Haggman phase shift addition [4] to the original Rapp equation, $\phi_{HH}(r)$, as shown in Table 1, is noted to have a descending phase with increasing input level but with a maximum of zero degrees phase shift. In order to accommodate devices that do not have a maximum of zero degrees phase shift, we propose an additional offset parameter, d , to the Honkanen & Haggman phase shift model. Hence the updated phase shift model $\phi_{HHO}(r)$ is given as equation (3);

$$\phi_{HHO}(r) = \frac{b}{1 + e^{-c(r-a)}} \cdot \frac{1}{p_1 r + p_2} - d \quad (3)$$

Where a is the cut-off input voltage at which the transistor begins to conduct, b is the switch-on step magnitude in degrees, c determines the steepness of the step-wise change, and p_1 and p_2 are parameters for adjusting the descending slope. The offset parameter d allows for positive phase shift. An example of the application is shown in Fig. 6 for the LDMOS 2-stage power amplifier [8].

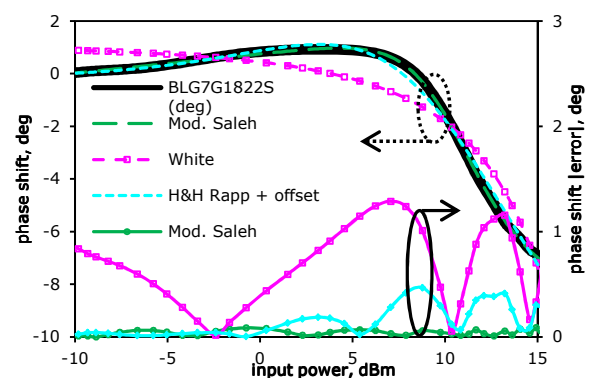


Fig. 6 AM/PM Curve fit of BLM7G1822S comparisons of O'Dorma et al, White et al & Honkanen & Haggman AM/PM

expression, with additional offset, as well as absolute phase error comparisons.

Even though the Honkanen & Haggman phase addition has coefficients that are related to bipolar transistor parameters; this model still performs well for LDMOS devices.

VI. CONCLUSION

This paper has demonstrated the need for fast system level comparisons to aid in the selection of critical RFPA output devices. For SSPA devices, having only below saturation data available, then the Original Rapp model [3], including the Honkanen & Haggman phase shift addition [4] plus the addition introduced in this paper, provides the best simple AM/AM and AM/PM model. If the available data extends into the saturation region then the O'Dorma et al [6] model provides the best behavioral model in that region.

This paper has also shown improvements for the use of two existing models plus provided an additional parameter, as well as providing a method for determining the saturation power from manufacturer's data. It has also compared the performance of these models when used beyond the provided manufacturer's data information, where typical modern digital modulation signals extend, thus resulting in further improvements and accuracies.

These simple model improvements and additions, combined with the extension of the modeling into the saturation region, thus allow for faster and more accurate system level analysis enabling better crucial output device selections to be made.

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REFERENCES

- [1] A. A. M. Saleh, "Frequency independent and frequency dependent non-linear models of TWT amplifiers", *IEEE Trans. Commun.*, vol. 29, no. 11, pp. 1715 – 1720, Nov. 1981.
- [2] A. Ghorbani and M. Sheikhan, "The effects of solid state power amplifier (SSPA) nonlinearities on M-PSK and M-QAM signal transmission", *6th Int. Conf. on Digital Processing of Signals in Communications, Loughborough, UK*, 1991, pp. 193 – 197.
- [3] C. Rapp, "Effects of HPA-nonlinearity on a 4-DPSK/OFDM-signal for a digital sound broadcasting system", *Proc. of 2nd European Conf. on Satellite communications, Liege, Belgium*, pp. 179 – 184, October 1991.
- [4] M. Honkanen and S-G. Haggman, "New Aspects on nonlinear power amplifier modelling in radio communication system simulations", *IEEE Commun., Lett.*, vol. 13, no. 6, pp. 844 – 848, June 1997.
- [5] G. P. White, A. G., Burr, and T., Javornik, "Modelling of nonlinear distortion in broadband fixed wireless access systems", *Electron. Lett.*, 2003, vol. 39, pp. 686 – 687, April 2003.
- [6] M. O'Dorma, S., Meza and Y. Lei, "New Modified Saleh Models for Memoryless Nonlinear Power Amplifier Behavioral Modelling", *IEEE Commun. Lett.*, vol. 13, No. 6, pp. 399 - 401, June 2009.
- [7] D. Schreurs, M., O'Dorma, A. A., Goacher, and M. Gadringer, "RF Power Amplifier Behavioral Modeling" (The Cambridge RF and Microwave Engineering series, Cambridge University Press, 2009, 1st edition), Ch. 3, Sec. 3.3, pp. 98 - 99.
- [8] NXP Semiconductors BLM7G1822S Objective data sheet', http://www.nxp.com/documents/data_sheet/BLM7G1822S-40PB_S-40PBG.pdf, accessed August 2014.